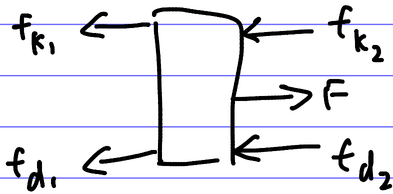
1)



$f_{k1} = k_1 x$ $\qquad$ $f_{k_2} = k_2 x$

$f_{d_1} = c_1 \dot{x}$ $\qquad$ $f_{d_2} = c_2 \dot{x}$

$$F = f_{k_1} + f_{k_2} + f_{d_1} + f_{d_2}$$

$$m\ddot{x} + (k_1 + k_2)x + (c_1 + c_2)\dot{x} = F$$

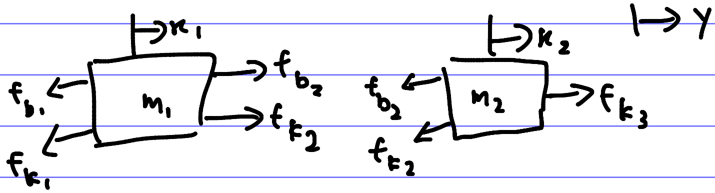
$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + (c_1+c_2)s + (k_1+k_2)}$$

2) $$k(y-x) = m\ddot{x}$$

$$m\ddot{x} + kx = ky$$

$$\frac{X(s)}{Y(s)} = \frac{k}{ms^2+k}$$

3)



$f_{b_1} = b_1\dot{x}_1$ $\quad f_{b_2} = b_2(\dot{x}_2 - \dot{x}_1)$ $\quad f_{k_3} = k_3(y - x_2)$

$f_{k_1} = k_1 x_1$ $\quad f_{k_2} = k_2(x_2 - x_1)$

$$m_1\ddot{x}_1 = b_2(\dot{x}_2 - \dot{x}_1) + k_2(x_2 - x_1) - b_1\dot{x}_1 - k_1\dot{x}_1$$
$$m_1\ddot{x}_1 + (b_1+b_2)\dot{x}_1 + (k_1+k_2)x_1 = b_2\dot{x}_2 + k_2x_2 \quad - (1)$$

$$m_2\ddot{x}_2 = k_3(y - x_2) - b_2(\dot{x}_2 - \dot{x}_1) - k_2(x_2 - x_1)$$
$$m_2\ddot{x}_2 + b_2 x_2 + (k_2+k_3)x_2 = b_2\dot{x}_1 + k_2x_1 + k_3y \quad - (2)$$

Applying Laplace Transform to (1) and (2):

$$[m_1s^2 + (b_1+b_2)s + (k_1+k_2)]X_1(s) = (b_2s + k_2)X_2(s)$$
$$[m_2s^2 + b_2s + (k_2+k_3)s]X_2(s) = (b_2s + k_2)X_1(s) + k_3Y(s)$$

4a) $J\frac{d\theta^2}{dt^2} + B\frac{d\theta}{dt} = \tau$

$J\dot{\omega} + B\omega = \tau$

$Js\,\Omega(s) + B\,\Omega(s) = T(s)$

$$\frac{\Omega(s)}{T(s)} = \frac{1}{10+100s}$$

b) $\tau = 100$

$T(s) = \frac{100}{s}$

$$\Omega(s) = \frac{1}{10+100s}\left(\frac{100}{s}\right)$$

$$= \frac{1}{0.1+s}\left(\frac{1}{s}\right)$$

$$= \frac{10}{s} - \frac{10}{s+0.1}$$

$$\omega(t) = 10 - 10e^{-0.1t}$$

$$\lim_{t\to\infty} \omega(t) = \lim_{t\to\infty} 10 - 10e^{-0.1t}$$

$$= 10 \text{ rads}^{-1}$$